



AI and ML

Tutorial 1

Student Id : 2417514

Student Name : Suvina Thapa

Section : L6CG18

Tutor : Shiv Kumar Yadav

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Tutorial 1

Question 1: Practice Problem on Preliminary Set Concepts:

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1.4) 1) Suppose that:

$A = \{x : x \in \mathbb{N} \text{ and } x \text{ is even}\},$
 $B = \{x : x \in \mathbb{N} \text{ and } x \text{ is prime}\}, \text{ and}$
 $C = \{x : x \in \mathbb{N} \text{ and } x \text{ is a multiple of } 5\}.$

a) $A \cap B$
→ Numbers which are both even and prime

b) $B \cap C$
→ Numbers which are both prime and multiple of 5

c) $A \cup B$
→ Numbers which are even OR prime. (Union)

d) $A \cap (B \cup C)$
→ $B \cup C$ - both Prime OR multiple of 5 then common with even numbers.

2) Prove:

i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

→ Here;

- First Inclusion: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Taking the left side:

$x \in A \cup (B \cap C)$ which means,
 $x \in A$ OR $x \in B \cap C$

Case 1: $x \in A$
 x is in $A \cup B$ and $A \cup C$ (both sets)
 $\therefore x \in (A \cup B) \cap (A \cup C)$

Case 2: $x \in B \cap C$ ($x \in B$ and $x \in C$)
 x is in $A \cup B$ and x is in $A \cup C$ (both set)
So, $x \in (A \cup B) \cap (A \cup C)$

$\therefore A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$

• Second Inclusion: $(A \cup B) \cap (A \cup C) = A \cup (B \cap C)$

Taking the left side:

$x \in (A \cup B) \cap (A \cup C)$ which means
 $x \in (A \cup B)$ and $x \in A \cup C$

Case 1: $x \in A \cup B$ / $x \in A$

x is in $A \cup (B \cap C)$

Case 2: $x \notin A$ / $x \in B$

~~x is in~~ $x \in A \cup B$ and $x \notin A$ it must
be $x \in B$

Similarly, since $x \in A \cup C$ and $x \notin A$ it must
be $x \in C$

$\therefore x \in B \cap C$. So, $x \in A \cup (B \cap C)$

Since, both cases imply that $x \in A \cup (B \cap C)$
which shown that $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$.

Conclusion

From Both first and second inclusion, we can
conclude that:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

→ Here;

• First Inclusion: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Taking the left side:

$x \in A \cap (B \cup C)$ which means

$x \in A$ and $x \in B \cup C$

Case 1: $x \in B$

x is in $A \cap B$

Case 2: $x \in C$

x is in $A \cap C$

$\therefore x \in (A \cap B) \cup (A \cap C)$

Thus,

$$A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$$

- Second Inclusion: $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$

Taking the left side:

$$(A \cap B) \cup (A \cap C)$$

Case 1: $x \in A \cap B$

So, x is in A and B which for $x \in B, x \in B \cup C$

$$\therefore x \in A \cap (B \cup C)$$

Case 2: $x \in A \cap C$

So, $x \in A, x \in C$

Thus, x is in $x \in B \cup C$

$$\therefore x \in A \cap (B \cup C)$$

Conclusion:

From both inclusion, we can conclude that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

3) Define a function: $f: \mathbb{N} \rightarrow \mathbb{N}$

a) that is one-to-one function

$$\rightarrow f(n) = 2n$$

Produce only even number not all natural number

b) that is onto but not one-to-one

$\rightarrow$ The function is onto if every element in

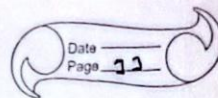
codomain has a corresponding element in domain

Here, the codomain is $\mathbb{N}$ However $f(n)$ only

produce even number so no $n \in \mathbb{N}$ mapping to

odd natural number.

Question2 : Matrix Operation and System of Linear Equations.



2.1 Matrix Operations and system of linear Eqⁿ

1(a) Write down 3x3 matrix with one on diagonal & zero

→ Required matrix:
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b) Write down 3x4 matrix with one in diagonal & zero

→ Required matrix:
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

c) Write down 4x3 matrix with one on diagonal & 0

→ Required matrix:
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

2. let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ Compute AD and DA

→ Here,

$$AD = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 2 + 1 \times 0 + 1 \times 0 & 1 \times 0 + 1 \times 3 + 1 \times 0 & 1 \times 0 + 1 \times 0 + 1 \times 4 \\ 1 \times 2 + 2 \times 0 + 3 \times 0 & 1 \times 0 + 2 \times 3 + 3 \times 0 & 1 \times 0 + 2 \times 0 + 3 \times 4 \\ 1 \times 2 + 3 \times 0 + 4 \times 0 & 1 \times 0 + 3 \times 3 + 4 \times 0 & 1 \times 0 + 3 \times 0 + 4 \times 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3 & 4 \\ 2 & 6 & 12 \\ 2 & 9 & 16 \end{bmatrix}$$

$$\begin{aligned}
 DA &= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix} \\
 &= \begin{bmatrix} 2 \times 1 + 0 \times 1 + 0 \times 1 & 0 \times 1 + 0 \times 2 + 0 \times 3 & 2 \times 1 + 0 \times 3 + 0 \times 1 \\ 0 \times 1 + 3 \times 1 + 0 \times 1 & 0 \times 1 + 3 \times 2 + 0 \times 3 & 0 \times 1 + 3 \times 3 + 0 \times 1 \\ 0 \times 1 + 0 \times 1 + 4 \times 1 & 0 \times 1 + 0 \times 2 + 4 \times 3 & 0 \times 1 + 0 \times 3 + 4 \times 4 \end{bmatrix} \\
 &= \begin{bmatrix} 2 & 2 & 2 \\ 3 & 6 & 9 \\ 4 & 12 & 16 \end{bmatrix}
 \end{aligned}$$

From observation,

$AD \neq DA$, So matrix multiplication is not commutative as D is a diagonal matrix

3. let $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix}$

Verify that $AB = AC$

→ Here,

$$AB = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 2 + 2 \times 1 & 1 \times 1 + 2 \times 3 \\ 2 \times 2 + 4 \times 1 & 2 \times 1 + 4 \times 3 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$$

$$AC = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 4 + 2 \times 0 & 1 \times 3 + 2 \times 2 \\ 2 \times 4 + 4 \times 0 & 2 \times 3 + 4 \times 2 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$$

$$\therefore AB = AC = \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$$

4. Prove that: $(AB)^T = B^T A^T$

→ Here,

→ By definition,

$$C = AB$$

$$C_{ij} = \sum_k a_{ik} b_{kj}$$

Transpose:

$$(AB)^T_{ij} = C_{ji}$$

$$C_{ji} = \sum_k a_{jk} b_{ki}$$

let,

A be an $m \times n$ matrix

B be an $n \times p$ matrix

Then,

AB is an $m \times p$ matrix

Transpose:

$$(AB)^T_{ij} = (AB)_{ji}$$

But,

$$(AB)_{ji} = \sum_{k=1}^n a_{jk} b_{ki}$$

So,

$$(AB)^T_{ij} = \sum_{k=1}^n a_{jk} b_{ki} \rightarrow \text{eq}^n \text{ci}$$

now, computing $B^T A^T$

For A:

$$(A^T)_{kj} = a_{jk}$$

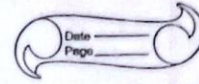
For B:

$$(B^T)_{ik} = b_{ki}$$

Multiplying $B^T A^T$

Then ij element of $B^T A^T$ is

$$(B^T A^T)_{ij} = \sum_{k=1}^n (B^T)_{ik} (A^T)_{kj}$$



Substituting transpose definitions:

$$(B^T)_{ik} = b_{ki}$$

$$(A^T)_{kj} = a_{jk}$$

$$\text{so, } (B^T A^T)_{ij} = \sum_{k=1}^n b_{ki} a_{jk} \rightarrow \text{eq}^n \text{ (i)})$$

(Computing both expression (i) and (ii))

both eqⁿ are commutative

$$a_{jk} b_{ki} = b_{ki} a_{jk}$$

Thus,

$$(AB)^T_{ij} = (B^T A^T)_{ij}$$

Hence proved.

5. $A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$ show that is a symmetric square matrix. Observe sum of $A^T A$ is sum of squares of elements of A .

→ Here,

$$\text{Transpose of } A (A^T) = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

$$A^T A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$$

$$= \begin{bmatrix} axa+bx b+cx c & axd+bx e+cx f \\ dx a+ex b+fx c & dx d+ex e+fx f \end{bmatrix}$$

$$= \begin{bmatrix} a^2+b^2+c^2 & ad+be+cf \\ ad+be+cf & d^2+e^2+f^2 \end{bmatrix}$$

This matrix is symmetric.

6. Find inverse of matrix $A = \begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix}$ $B = \begin{bmatrix} 6 & 4 \\ 3 & 3 \end{bmatrix}$

→ Here, Given

$$A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{\det(A)} \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

now,

$$A = \begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix}$$

$$|A| = 5 \times 5 - 6 \times 4 = 1$$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix}$$

$$|B| = 6 \times 3 - 4 \times 3 = 6$$

$$B^{-1} = \frac{1}{6} \begin{bmatrix} 3 & -4 \\ -3 & 6 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{2}{3} \\ -\frac{1}{2} & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix} \text{ and } B^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{2}{3} \\ -\frac{1}{2} & 1 \end{bmatrix}$$

7. Given matrices A and B, verify they are inverse

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}$$

→ Here;

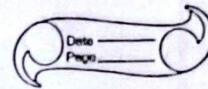
Two matrices A and B are inverse if

$$AB = I \text{ and } BA = I \text{ where } I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

now,

$$AB = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} 4 \times 1 + 1 \times (-3) & 4 \times (-1) + 1 \times 4 \\ 3 \times 1 + 1 \times (-3) & 3 \times (-1) + 1 \times 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$BA = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 \times 4 + (-1) \times 3 & -1 \times 1 + (-1) \times 1 \\ (-3) \times 4 + 4 \times 3 & (-3) \times 1 + 4 \times 1 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{now, } AB = BA = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

8. Verify that given matrices are inverse of each other

$$\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix} : \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix}$$

→ Here;

$$\text{let } A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix}$$

To be inverse of each other,

$$AB = BA = I \text{ where } I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

now,

$$AB = \begin{bmatrix} 1 \times 3 + (-1) \times 2 + 0 \times 3 & 1 \times (-4) + (-1) \times (-4) + 0 \times (-5) & 1 \times 1 + (-1) \times 1 + 0 \times 1 \\ 1 \times 3 + 0 \times 2 + (-1) \times 3 & 1 \times (-4) + 0 \times (-4) + (-1) \times (-5) & 1 \times 1 + 0 \times 1 + (-1) \times 1 \\ 2 \times 3 + 3 \times 2 + (-4) \times 3 & 2 \times (-4) + 3 \times (-4) + (-4) \times (-5) & 2 \times 1 + 3 \times 1 + (-4) \times 1 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Since,

$$AB = BA = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{Proved}$$

3.4. (1) System with Two Variable

a)
$$\begin{cases} x + 2y = 4 \\ 3x - 5y = 1 \end{cases}$$

→ Here, Given:

$$x + 2y = 4 \longrightarrow \text{eq}^n \text{ (i)}$$

$$3x - 5y = 1 \longrightarrow \text{eq}^n \text{ (ii)}$$

now,

Converting into Matrix Form

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \quad \text{which mean}$$

$$\therefore AX = B$$

$$\text{Determinant of A } (|A|) = 1 \times (-5) - 3 \times 2 = -5 - 6 = -11 \neq 0$$

we know,

$$A^{-1} = \frac{1}{|A|} \times \text{Adj of A}$$

$$|A| = -11$$

$$= \frac{1}{-11} \times \begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix}$$

$$\text{To find } X = A^{-1}B$$

$$= \frac{1}{-11} \begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$= \frac{1}{-11} \begin{bmatrix} -5 \times 4 - 2 \times 1 \\ -3 \times 4 + 1 \times 1 \end{bmatrix}$$

$$= \frac{1}{-11} \begin{bmatrix} -24 \\ -11 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\therefore x = 2 \text{ and } y = 1$$

b)
$$\begin{cases} 5x + y = 13 \\ 3x + 2y = 5 \end{cases}$$

→ Here: Given:

$5x + y = 13 \rightarrow \text{eq}^{\text{th}} \text{ci}$

$3x + 2y = 5 \rightarrow \text{eq}^{\text{th}} \text{cii}$

now,

Converting into Matrix form

$A = \begin{bmatrix} 5 & 1 \\ 3 & 2 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$, $B = \begin{bmatrix} 13 \\ 5 \end{bmatrix}$ which is $AX = B$

Determinant of A ($|A|$) = $5 \times 2 - 1 \times 3 = 7$

we know,

$A^{-1} = \frac{1}{|A|} \times \text{Adj of } A$

$= \frac{1}{7} \times \begin{bmatrix} 2 & -1 \\ -3 & 5 \end{bmatrix}$

To find $X = A^{-1}B$

$= \frac{1}{7} \begin{bmatrix} 2 & -1 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 13 \\ 5 \end{bmatrix}$

$= \frac{1}{7} \begin{bmatrix} 2 \times 13 - 1 \times 5 \\ -3 \times 13 + 5 \times 5 \end{bmatrix}$

$= \frac{1}{7} \begin{bmatrix} 21 \\ -14 \end{bmatrix}$

$= \begin{bmatrix} 3 \\ -2 \end{bmatrix}$

$\therefore x = 3 \text{ and } y = -2$

$$d) \begin{cases} 2x + 3y = 6 \\ x - y = \frac{1}{2} \end{cases}$$

→ Here, Given

$$2x + 3y = 6 \longrightarrow \text{eq (i)}$$

$$x - y = \frac{1}{2} \longrightarrow \text{eq (ii)}$$

now,

Converting into Matrix form

$$A = \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 6 \\ \frac{1}{2} \end{bmatrix} \text{ which is } AX = B$$

$$\text{Determinant of } A (|A|) = 2 \times 1 - 3 \times 1 = -5 \neq 0$$

we know,

$$A^{-1} = \frac{1}{|A|} \times \text{Adj of } A$$

$$= \frac{1}{-5} \times \begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix}$$

To find $X = A^{-1}B$

$$= \frac{1}{-5} \begin{bmatrix} -1 & -3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ \frac{1}{2} \end{bmatrix}$$

$$= \frac{1}{-5} \begin{bmatrix} -1 \times 6 + (-3) \times \frac{1}{2} \\ -1 \times 6 + 2 \times \frac{1}{2} \end{bmatrix}$$

$$= \frac{1}{-5} \begin{bmatrix} -6 - \frac{3}{2} \\ -6 + 1 \end{bmatrix}$$

$$= \frac{1}{-5} \begin{bmatrix} -6 - \frac{3}{2} \\ -6 + 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{2} \\ 1 \end{bmatrix}$$

$$\therefore x = \frac{3}{2} \text{ and } y = 1$$

e) $\begin{cases} 4x + 3y = 11 \\ x - 3y = -1 \end{cases}$

→ Here; Given

$4x + 3y = 11 \longrightarrow \text{eq}^n \text{ (i)}$

$x - 3y = -1 \longrightarrow \text{eq}^n \text{ (ii)}$

now,

Converting into Matrix form

$A = \begin{bmatrix} 4 & 3 \\ 1 & -3 \end{bmatrix}$ $X = \begin{bmatrix} x \\ y \end{bmatrix}$ $B = \begin{bmatrix} 11 \\ -1 \end{bmatrix}$ which is $AX = B$

Determinant of A ($|A|$) $= 4 \times (-3) - 3 \times 1 = -15 \neq 0$

we know,

$A^{-1} = \frac{1}{|A|} \times \text{Adj of } A$

$= \frac{1}{-15} \times \begin{bmatrix} -3 & -3 \\ -1 & 4 \end{bmatrix}$

To Find $X = A^{-1}B$

$= \frac{1}{-15} \begin{bmatrix} -3 & -3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 11 \\ -1 \end{bmatrix}$

$= \frac{1}{-15} \begin{bmatrix} (-3) \times 11 + 3 \times (-1) \\ (-1) \times 11 - 4 \times (-1) \end{bmatrix}$

$= \frac{1}{-15} \begin{bmatrix} -30 \\ -15 \end{bmatrix}$

$= \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$\therefore x = 2 \text{ and } y = 1$

Question3: Practice Problem on Solving System of Linear Equation using Matrix Method:

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2. System with Three Variables:

a)
$$\begin{cases} x+y-2 = 2 \\ x+2 = 7 \\ 2x+y+2 = 13 \end{cases}$$

→ Here,
Converting into augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 1 & 0 & 1 & 7 \\ 2 & 1 & 1 & 13 \end{array} \right] = [A|B]$$

Performing Elementary Row Operation until we reached Row Echelon Form:

- $R_2 \rightarrow R_2 - R_1$
 $= [1-1, 0-1, 1-(-1) | 7-2] = [0, -1, 2 | 5]$
- $R_3 \rightarrow R_3 - 2R_1$
 $= [2-2, 1-2, 1-(-2) | 13-4] = [0, -1, 3 | 9]$

new matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & 2 & 5 \\ 0 & -1 & 3 & 9 \end{array} \right]$$

- $R_3 \rightarrow R_3 - R_2$
 $= [0-0, -1-(-1), 3-2 | 9-5] = [0, 0, 1 | 4]$

new matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & 2 & 5 \\ 0 & 0 & 1 & 4 \end{array} \right]$$

now, Row Echelon Form:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & 2 & 5 \\ 0 & 0 & 1 & 4 \end{array} \right] \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix} \text{ --- eqn (i)}$$

Writing eqⁿ (i) as linear combination

$$z \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

which in eqⁿ is $\begin{cases} x+y-2=2 \\ -y+2z=5 \\ z=4 \end{cases} \rightarrow \text{eqⁿ (ii)}$

$$\therefore z = 4$$

From eqⁿ (ii) $-y+2z=5$

$$\text{or, } -y+2 \times 4 = 5$$

$$\text{or, } -y = 5-8$$

$$\therefore y = 3$$

From eqⁿ (iii) $x+y-2=2$

$$\text{or, } x+3-4=2$$

$$\therefore x = 3$$

$$\therefore x=3, y=3 \text{ and } z=4$$

b) $\begin{cases} x+y-2=2 \\ 3x+y=7 \\ x+y+2z=3 \end{cases}$

- Here;

Writing in Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 3 & 1 & 0 & 7 \\ 1 & 1 & 2 & 3 \end{array} \right] = [A|B]$$

Performing Elementary Row Operations

$$\bullet R_2 \rightarrow R_2 - 3R_1$$

$$= [3-3, 1-3, 0-(-3) | 7-6] = [0, -2, 3 | 1]$$

$$\bullet R_3 \rightarrow R_3 - R_1$$

$$= [1-1, 1-1, 2-(-1) | 3-2] = [0, 0, 3 | 1]$$

new matrix X:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -2 & 3 & 1 \\ 0 & 0 & 3 & 1 \end{array} \right] \rightarrow (i)$$

Writing matrix (i) as linear combination

$$x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{which in eq}^n \text{ is } \begin{cases} x + y - 2 = 2 \\ -2y + 3z = 1 \\ 3z = 1 \end{cases} \text{ - (ii)}$$

$$3z = 1$$

$$\therefore z = \frac{1}{3}$$

From eqⁿ (ii),

$$-2y + 3z = 1 \text{ and } x + y - 2 = 2$$

$$\text{or } -2y + 1 = 1 \quad \text{or } x + 0 - \frac{1}{3} = 2$$

$$\therefore y = 0$$

$$\therefore x = \frac{7}{3}$$

$$\therefore x = \frac{7}{3}, y = 0 \text{ and } z = \frac{1}{3}$$

$$c) \begin{cases} -x - 2y + z = 1 \\ 2x + 3y = 2 \\ y - 2z = 0 \end{cases}$$

→ Here:

Writing into Augmented Matrix:

$$\left[\begin{array}{ccc|c} -1 & -2 & 1 & 1 \\ 2 & 3 & 0 & 2 \\ 0 & 1 & -2 & 0 \end{array} \right] = [A|B]$$

Performing Elementary Row Operation

• $R_1 = R_1 / -1$

$= \left[-\frac{1}{-1}, -\frac{2}{-1}, \frac{1}{-1} \mid \frac{1}{-1} \right] = [1, 2, -1 \mid -1]$

• new matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 2 & 3 & 2 & 2 \\ 0 & 1 & -2 & 0 \end{array} \right]$$

• $R_2 = R_2 - 2R_1$

$= [2-2, 3-4, 0-(-2) \mid 2-(-2)] = [0, -1, 2 \mid 4]$

new matrix

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & -1 & 2 & 4 \\ 0 & 1 & -2 & 0 \end{array} \right]$$

• $R_3 = R_3 + R_2$

$= [0+0, 1+(-1), -2+2 \mid 0+4] = [0, 0, 0 \mid 4]$

new matrix

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & -1 & 2 & 4 \\ 0 & 0 & 0 & 4 \end{array} \right]$$

this matrix has no solution.

d)
$$\begin{cases} x+4y-2=4 \\ 2x+5y+8z=15 \\ x+3y-3z=1 \end{cases}$$

→ Here,

Writing matrix equation augmented

$$AX=B \quad [A|I] = \left[\begin{array}{ccc|ccc} 1 & 4 & -1 & 1 & 0 & 0 \\ 2 & 5 & 8 & 0 & 1 & 0 \\ 1 & 3 & -3 & 0 & 0 & 1 \end{array} \right]$$

Performing Elementary Row Operations

• $R_2 = R_2 - 2R_1$

$$= [2-2 \times 1, 5-2 \times 4, 8-2 \times (-1) \mid 0-2 \times 1, 1-2 \times 0, 0-2 \times 0]$$

$$= [0, -3, 10 \mid -2, 1, 0]$$

• $R_3 \rightarrow R_3 - R_1$

$$= [1-1, 3-4, -3-(-1) \mid 0-1, 0-0, 1-0]$$

$$= [0, -1, -2 \mid -1, 0, 1]$$

new matrix

$$\left[\begin{array}{ccc|ccc} 1 & 4 & -1 & 1 & 0 & 0 \\ 0 & -3 & 10 & -2 & 1 & 0 \\ 0 & -1 & -2 & -1 & 0 & 1 \end{array} \right]$$

• $R_2 \rightarrow R_2$
 $\frac{-3}{-3}$

$$= [0, 1, -10/3 \mid -2/3, 1/3, 0]$$

new matrix

$$\left[\begin{array}{ccc|ccc} 1 & 4 & -1 & 1 & 0 & 0 \\ 0 & 1 & -10/3 & -2/3 & 1/3 & 0 \\ 0 & -1 & -2 & -1 & 0 & 1 \end{array} \right]$$

• $R_1 \rightarrow R_1 - 4 \times R_2$

$$= [1, 0, -1-4 \times (-10/3), \mid 1-4 \times (-2/3), 0-4 \times (1/3), 0]$$

$$= [1, 0, 37/3, \mid 5/3, -4/3, 0]$$

• $R_3 \rightarrow R_3 + R_2$

$= [0, 0, -2 + (-10/3) \mid -1 + 2/3, 0 + (-1/3), 1 + 0]$

$= [0, 0, -16/3 \mid -1/3, -1/3, 1]$

new matrix:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 37/3 & -5/3 & 4/3 & 0 \\ 0 & 1 & -10/3 & 2/3 & -1/3 & 0 \\ 0 & 0 & -16/3 & -1/3 & -1/3 & 1 \end{array} \right]$$

• $R_3 \rightarrow R_3 \times (-3/16)$

$= [0, 0, 1 \mid (-1/3) \times (-3/16), (-1/3) \times (-3/16), 1 \times (-3/16)]$

$= [0, 0, 1, 1/16, 1/16, -3/16]$

new matrix:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 37/3 & -5/3 & 4/3 & 0 \\ 0 & 1 & -10/3 & 2/3 & -1/3 & 0 \\ 0 & 0 & 1 & 1/16 & 1/16 & -3/16 \end{array} \right]$$

• $R_1 \rightarrow R_1 - 37/3 \times R_3$

• $R_2 \rightarrow R_2 + 10/3 \times R_3$

$= [1 - \frac{37}{3} \times 0, 0 - \frac{37}{3} \times 0, \frac{37}{3} - \frac{37}{3} \times 1 \mid -5/3 - \frac{37}{3} \times \frac{1}{16}, 4/3 - \frac{37}{3} \times \frac{1}{16}, 0 - \frac{37}{3} \times (-\frac{3}{16})]$

$= [1, 0, 0 \mid -117/48, 27/48, 11/48]$

$\rightarrow [0 + \frac{10}{3} \times 0, 1 + \frac{10}{3} \times 0, -\frac{10}{3} + \frac{10}{3} \times 1 \mid \frac{2}{3} + \frac{10}{3} \times \frac{1}{16}, -\frac{1}{3} + \frac{10}{3} \times \frac{1}{16}, 0 + \frac{10}{3} \times (-\frac{3}{16})]$

$= [0, 1, 0 \mid 42/48, -6/48, -30/48]$

Final new matrix:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -11/48 & 27/48 & 11/48 \\ 0 & 1 & 0 & 42/48 & -6/48 & -30/48 \\ 0 & 0 & 1 & 1/6 & 1/6 & -3/16 \end{array} \right]$$

Thus,

$$A^{-1} = \begin{bmatrix} -11/48 & 27/48 & 11/48 \\ 42/48 & -6/48 & -30/48 \\ 1/6 & 1/6 & -3/16 \end{bmatrix}$$

$$X = A^{-1}B$$

$$B = \begin{bmatrix} 4 \\ 15 \\ 1 \end{bmatrix}$$

Finding value of x :

$$x = \frac{-11}{48} \times 4 + \frac{27}{48} \times 15 + \frac{11}{48} \times 1$$

$$\therefore x = 3$$

$$y = \frac{42}{48} \times 4 + \left(\frac{-6}{48} \right) \times 15 + \left(\frac{-30}{48} \right) \times 1$$

$$\therefore y = 1$$

$$z = \frac{1}{16} \times 4 + \frac{1}{16} \times 15 + \left(\frac{-3}{16} \right) \times 1$$

$$\therefore z = -1$$

$$\therefore x = 3, y = 1 \text{ and } z = -1$$

Question4: Bonus Problem

Date 29
 Page 29

3. Bonus Problem

1. Confirm why the following has no solution:

$$\begin{cases} x + 2y - 3z = -1 \\ x - 3y + 2z = 1 \\ 2x - y - 3z = 2 \end{cases}$$

> Here,

Writing into augmented matrix :

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 1 & -3 & 2 & 1 \\ 2 & -1 & -3 & 2 \end{array} \right]$$

- $R_2 \rightarrow R_2 - R_1$
 $= [1, -3, 2, 1] - [1, 2, -3, -1]$
 $= [0, -5, 4, 2]$
- $R_3 \rightarrow R_3 - 2R_1$
 $= [2, -1, -3, 2] - [2, 4, -6, -2]$
 $= [0, -5, 3, 4]$

new matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & -5 & 4 & 2 \\ 0 & -5 & 3 & 4 \end{array} \right]$$

- $R_3 \rightarrow R_3 - R_2$
 $= [0, -5, 3, 4] - [0, -5, 4, 2]$
 $= [0, 0, -1, 2]$

new matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & -5 & 4 & 2 \\ 0 & 0 & -1 & 2 \end{array} \right]$$

Linear equation

$$-2 = 2$$

$$\therefore 2 = -2,$$

$$-5y + 4z = 2 \quad \text{and} \quad x + 2y - 3z = -1$$

$$\text{or } -5y + 4 \times (-2) = 2$$

$$\text{or } x + 2 \times (-2) - 3 \times (-2) = -1$$

$$\text{or } -5y - 8 = 2$$

$$\text{or } x - 4 + 6 = -1$$

$$\therefore y = -2$$

$$\text{or } x + 2 = -1$$

$$\therefore x = -3$$

$\therefore$ So, the system has a solution.

2 b) Confirm why the following system has no solution

$$\begin{cases} 2x - 2y + 3z = 6 \\ 4x - 3y + 2z = 0 \\ -2x + 3y - 7z = 1 \end{cases}$$

$\rightarrow$ Here,

Convert into Augmented Matrix

$$\left[\begin{array}{ccc|c} 2 & -2 & 3 & 6 \\ 4 & -3 & 2 & 0 \\ -2 & 3 & -7 & 1 \end{array} \right]$$

making pivot 1

$$R_1 \rightarrow R_1/2$$

$$= \left[\begin{array}{ccc|c} 1 & -1 & 3/2 & 3 \\ 4 & -3 & 2 & 0 \\ -2 & 3 & -7 & 1 \end{array} \right] = [1, -1, 3/2, 3]$$

$$R_2 \rightarrow R_2 - 4R_1$$

$$= \left[\begin{array}{ccc|c} 1 & -1 & 3/2 & 3 \\ 0 & 1 & -4 & -12 \\ -2 & 3 & -7 & 1 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & -5 & -9 \\ 0 & 1 & -4 & -12 \\ -2 & 3 & -7 & 1 \end{array} \right]$$

• $R_3 \rightarrow R_3 + 2R_1$

= $[0, 1, -4, 7]$

new matrix:

$$\left[\begin{array}{ccc|c} 1 & -1 & 3/2 & 3 \\ 0 & 1 & -4 & -12 \\ 0 & 1 & -4 & 7 \end{array} \right]$$

$R_3 \rightarrow R_3 - R_2$

= $[0, 0, 0, 19]$

new matrix:

$$\left[\begin{array}{ccc|c} 1 & -1 & 3/2 & 3 \\ 0 & 1 & -4 & -12 \\ 0 & 0 & 0 & 19 \end{array} \right]$$

now,

$$0x + 0y + 0z = 19$$

$$0 = 19$$

∴ The system has no solution.

3. Confirm why the following system has infinitely many solutions:

$$\begin{cases} x + 2y - z = 1 \\ 2x + 7y + 4z = 11 \\ x + 3y + 2z = 4 \end{cases}$$

→ Here,

converting into Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 7 & 4 & 11 \\ 1 & 3 & 2 & 4 \end{array} \right]$$

$$\begin{aligned} & \bullet R_2 \rightarrow R_2 - 2R_1 \\ & = [2 \ 7 \ 4 \ | \ 11] - 2[1 \ 2 \ -1 \ | \ 1] \\ & = [2 \ 7 \ 4 \ | \ 11] - [2 \ 4 \ -2 \ | \ 2] \\ & = [0 \ 3 \ 6 \ | \ 9] \end{aligned}$$

$$\begin{aligned} & \bullet R_3 \rightarrow R_3 - R_1 \\ & = [1 \ 3 \ 1 \ | \ 4] - [1 \ 2 \ -1 \ | \ 1] \\ & = [1-1 \ 3-2 \ 1-(-1) \ | \ 4-1] \\ & = [0 \ 1 \ 2 \ | \ 3] \end{aligned}$$

new matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 3 & 6 & 9 \\ 0 & 1 & 2 & 3 \end{array} \right]$$

$$\begin{aligned} & \bullet R_2 \rightarrow R_2 - 3R_3 \\ & = [0 \ 0 \ 0 \ | \ 0] \end{aligned}$$

new matrix

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 \end{array} \right] \quad \text{(Now swap } R_2 \leftrightarrow R_3)$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

now,

from 3rd row, we can conclude,

$$0x + 0y + 0z = 0$$

$$0 = 0$$

The system has many solutions.

Question5: Problem on derivative

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4.1.1 Find $f'(x)$

2) $f(x) = 2 + \frac{1}{x} + \frac{1}{x^2}$
 - Here, Given;
 using power
 $f(x) = 2 + x^{-1} + x^{-2}$
 now, formula: $\frac{d}{dx} x^n = nx^{n-1}$
 $f'(x) = 0 - x^{-2} - 2x^{-3}$
 $= -\frac{1}{x^2} - \frac{2}{x^3}$

3) $f(x) = \sqrt{x} \cos x$
 - Here, Given
 let, $u = \sqrt{x}$, $v = \cos x$
 now,
 Find derivative
 $f'(x) = \frac{d}{dx} \sqrt{x} + \frac{d}{dx} \cos x$
 using product rule
 $(uv)' = u'v + uv'$
 then,
 $u' = \frac{d}{dx} (x^{1/2}) = \frac{1}{2} x^{-1/2}$
 $v' = \frac{d}{dx} (\cos x) = -\sin x$
 using product rule:
 $f'(x) = \frac{1}{2} x^{-1/2} \cos x + x^{1/2} (-\sin x)$
 $= \frac{1}{2} x^{-1/2} \cos x - x^{1/2} \sin x$
 $= \frac{\cos x}{2\sqrt{x}} - \sqrt{x} \sin x$

4) $f(x) = x^3 e^x$

→ Here;

$$f'(x) = \frac{d}{dx} (x^3 e^x)$$

Using product rule

let $u = x^3$, $v = e^x$

then,

$$u' = \frac{d}{dx} (x^3) = 3x^2$$

$$v' = \frac{d}{dx} (e^x) = e^x$$

now,

$$\begin{aligned} f'(x) &= 3x^2 e^x + x^3 e^x \\ &= e^x (3x^2 + x^3) \end{aligned}$$

5) $f(x) = x^3 e^x$

- soln:

$$f'(x) = \frac{d}{dx} \left(\frac{x^3}{e^x} \right)$$

Using quotient rule,

$$\left(\frac{u}{v} \right)' = \frac{vu' - uv'}{v^2}$$

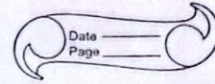
let,

$u = x^3$ and $v = e^x$

$$u' = \frac{d}{dx} (x^3) = 3x^2$$

$$v' = \frac{d}{dx} e^x = e^x$$

Applying Quotient Rule



$$\begin{aligned} f'(x) &= \frac{e^x(3x^2) - x^3(e^x)}{(e^x)^2} \\ &= \frac{(3x^2 - x^3)e^x}{e^{2x}} \\ &= \frac{3x^2 - x^3}{e^x} \end{aligned}$$

6) $f(x) = (3x^2 + 2x + 1)^5$

- Here;

$$f'(x) = \frac{d}{dx} (3x^2 + 2x + 1)^5$$

using chain rule

$$\frac{d}{dx} (g(x))^n = n(g(x))^{n-1} g'(x)$$

let,

$$g(x) = 3x^2 + 2x + 1$$

$$g'(x) = \frac{d}{dx} (3x^2 + 2x + 1)$$

$$= 6x + 2$$

Applying chain rule,

$$f'(x) = 5(3x^2 + 2x + 1)^4 (6x + 2)$$

7) $f(x) = (\sin x)^4$

- Here;

Using chain rule

$$f(x) = \sin^4 x$$

Outer func = u^4 = derivative $\rightarrow 4u^3$

Inner " = $u = \sin x \rightarrow$ " $\rightarrow \cos x$

$$f'(x) = 4(\sin x)^3 \cdot \cos x$$

8) $f(x) = e^{(3x^2 + 2x)}$

Here;

Using chain rule

$$f(x) = e^{g(x)} \rightarrow f'(x) = g'(x) e^{g(x)}$$

here,

$$g(x) = 3x^2 + 2x$$

$$g'(x) = 6x + 2$$

$$f'(x) = (6x + 2) e^{3x^2 + 2x}$$

9) $f(x) = \ln(5x^2 + 3x)$

Here;

Using chain rule

$$\frac{d}{dx} \ln(g(x)) = \frac{g'(x)}{g(x)}$$

now,

$$g(x) = 5x^2 + 3x$$

$$g'(x) = 10x + 3$$

$$f'(x) = \frac{10x + 3}{5x^2 + 3x}$$

10) $f(x) = \sqrt{x^3 + 4x + 1}$

Here;

$$f(x) = (x^3 + 4x + 1)^{1/2}$$

Using chain rule:

$$\text{Outer: } u^{1/2} \rightarrow \text{derivative} \rightarrow \frac{1}{2} u^{-1/2}$$

$$\text{Inner: } x^3 + 4x + 1 \rightarrow \text{derivative} \rightarrow 3x^2 + 4$$

$$f'(x) = \frac{1}{2} (x^3 + 4x + 1)^{-1/2} (3x^2 + 4)$$

$$f'(x) = \frac{3x^2 + 4}{2\sqrt{x^3 + 4x + 1}}$$

Question6: Bonus Problem on Gradients

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Bonus on Gradients

1. Evaluate the Gradient of a function.

i) $f(x, y) = x^2 + y^2$ at
(a) (0, 0) b) (1, 3) c) (-1, -5)

→ The gradient is:
$$\nabla f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$
$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 2y$$

a) At (0, 0)
$$\nabla f(0, 0) = (2 \times 0, 2 \times 0) = (0, 0)$$

b) At (1, 3)
$$\nabla f(1, 3) = (2 \times 1, 2 \times 3) = (2, 6)$$

c) At (-1, -5)
$$\nabla f(-1, -5) = (2 \times (-1), 2 \times (-5)) = (-2, -10)$$

ii) $f(x, y, z) = x^3 z - 2y^2 x + 5z$
→ The gradient is:
$$\nabla f(x, y, z) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

Partial derivatives

$$\frac{\partial f}{\partial x} = 3x^2z - 2y^2$$

$$\frac{\partial f}{\partial y} = -4xy$$

$$\frac{\partial f}{\partial z} = x^3 + 5$$

a) At $(1, 1, -4)$

→ $\nabla f(1, 1, -4)$

now,

$$\frac{\partial f}{\partial x} = 3 \times 1^2 \times (-4) - 2 \times 1^2 = -12 - 2 = -14$$

$$\frac{\partial f}{\partial y} = -4 \times 1 \times 1 = -4$$

$$\frac{\partial f}{\partial z} = 1^3 + 5 = 6$$

$$\nabla f(1, 1, -4) = (-14, -4, 6)$$

b) At $(0, 1, 0)$

$$\rightarrow \frac{\partial f}{\partial x} = 3 \times 0^2 \times 0 - 2 \times 1^2 = -2$$

$$\frac{\partial f}{\partial y} = -4 \times 0 \times 1 = 0$$

$$\frac{\partial f}{\partial z} = 0^3 + 5 = 5$$

$$\nabla f(0, 1, 0) = (-2, 0, 5)$$

c) At $(-3, -2, 1)$

$$\rightarrow \frac{\partial f}{\partial x} = 3x(-3)^2 \times 1 - 2 \times (-2)^2 = 27 - 8 = 19$$

$$\frac{\partial f}{\partial y} = -4 \times (-3) \times (-2) = -24$$

$$\frac{\partial f}{\partial z} = (-3)^3 + 5 = -27 + 5 = -22$$

$$\therefore \nabla f(-3, -2, 1) = (19, -24, -22)$$

Problem 2

A Suppose $g(x) = x^2 - 3x + 2$ and $f(x)$ is differentiable function: $f(0) = 3$ and $f'(0) = -1$, $f(1) = 5$ and $f'(1) = 0$, $f(2) = -3$ and $f'(2) = 3$, $f(3) = 6$ and $f'(3) = 1$

→

1. Find derivative of $f(x) \cdot g(x)$ at $x = 1$

- Here:

Using Product Rule:

$$(fg)' = f'g + fg'$$

$$f(1) = 5 \text{ and } f'(1) = 0$$

$$g(1) = 1^2 - 3 \cdot 1 + 2 = 0$$

$$g'(x) = 2x - 3$$

$$= 2 \cdot 1 - 3 = -1$$

Apply formula:

$$(fg)' = f'(1) \cdot g(1) + f(1) \cdot g'(1) \\ = 0 \cdot 0 + 5 \cdot (-1) = -5$$

2. Find derivative $\frac{f(x)}{g(x)}$ at $x = 0$

- Here:

Using Quotient Rule:

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

now,

$$\left(\frac{f}{g}\right)'(0) = \frac{(-1) \cdot 2 - 3 \cdot (-3)}{2^2} \\ = \frac{-2 + 9}{4} = \frac{7}{4}$$

Question 7: Practice Problem on Maxima and Minima using derivative

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4.2.1. Find maximum or minimum at interval

1) $g(x) = x^2 - 4x + 5$ at interval $[1, 4]$
 - Here;

$$g'(x) = \frac{d}{dx} (x^2 - 4x + 5) = 2x - 4$$
 If derivative = 0 (critical point)

$$2x - 4 = 0 \Rightarrow x = 2$$

$$g(2) = 2^2 - 4(2) + 5 = 4 - 8 + 5 = 1$$
 now,

$$g(1) = 1 - 4 + 5 = 2$$

$$g(4) = 16 - 16 + 5 = 5$$
 $\therefore$ Minimum at $x = 2$ and maximum at $x = 4$

2) $h(x) = \frac{x^2}{2} - 2x$ at interval $[-2, 3]$
 - Here;

$$h'(x) = \frac{d}{dx} \left(\frac{x^2}{2} - 2x \right) = x - 2$$
 Critical point

$$x - 2 = 0 \Rightarrow x = 2$$

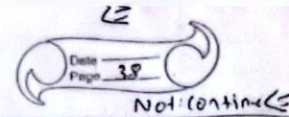
$$h(2) = \frac{2^2}{2} - 2(2) = \frac{4}{2} - 4 = 2 - 4 = -2$$
 At intervals

$$h(-2) = \frac{(-2)^2}{2} - 2(-2) = \frac{4}{2} + 4 = 6$$

$$h(3) = \frac{9}{2} - 6 = 4.5 - 6 = -1.5$$
 $\therefore$ minimum $(h(2)) = -2$
 and maximum at $x(-2) = 6$

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3) $p(x) = 3x^3 - 9x^2 + 7$ at interval $[-1, 2]$

→ Here,

$$p'(x) = 9x^2 - 18x = 9x(x-2)$$

Critical point

$$9x(x-2) = 0 \Rightarrow x = 0, 2$$

$$p(0) = 3(0) - 9(0) + 7 = 7$$

$$p(2) = 3(8) - 9(4) + 7 = -5$$

now,

At intervals

$$p(-1) = 3(-1) - 9(1) + 7 = -3 - 9 + 7 = -5$$

∴ Maximum at $x = 0$ which is 7 and minimum at $x = -1$ and 2 which is -5.

4) $f(x) = x^2 - 4x + 6$ at interval $[0, 3]$

→ Here,

$$f'(x) = 2x - 4 = 2(x-2)$$

Critical point

$$2x - 4 = 0 \Rightarrow x = 2$$

$$f(2) = 4 - 8 + 6 = 2$$

At interval

$$f(0) = 0 - 0 + 6 = 6$$

$$f(3) = 9 - 12 + 6 = 3$$

∴ Maximum at $x = 0$ which is 6 and min at $x = 2$

5) $q(x) = -2x^3 + 3x^2 + 12x$ at interval $[-1, 2]$

→ Here,

$$q'(x) = -6x^2 + 6x + 12 = -6x^2 + 6x + 12$$

$$\text{Critical point: } -6x^2 + 6x + 12 = 0 \Rightarrow x = 2, x = -1$$

$$q(2) = -16 + 12 + 24 = 20$$

$$q(-1) = 2 + 3 - 12 = -7$$

$$\text{At Interval: } q(-1) = 2 + 3 - 12 = -7$$

∴ Maximum at $x = 2$ which is 20 and minimum at $x = -1$ which is -7.

Question8: Problem: Using the Gradient to Evaluate Maxima and Minima of a Multivariate Function over a Specific Interval: Problem: Using the Gradient to Evaluate Maxima and Minima of a Multivariate Function over a Specific Interval:

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2. Using Gradient to evaluate Maxima and Minima

1. $m(x, y) = -x^2 - y^2 + 4x + 6y$ interval $x: [0, 4]$ & $y: [0, 5]$

→ Here

Gradient:

$$\frac{\partial m}{\partial x} = -2x + 4, \quad \frac{\partial m}{\partial y} = -2y + 6$$

$$\nabla m = \begin{bmatrix} -2x + 4 \\ -2y + 6 \end{bmatrix}$$

let gradient = 0 (critical point)

$$-2x + 4 = 0 \Rightarrow x = 2$$

$$-2y + 6 = 0 \Rightarrow y = 3$$

∴ critical point = (2, 3)

$$m(2, 3) = -(2)^2 - (3)^2 + 4(2) + 6(3)$$

$$= -4 - 9 + 8 + 18 = 13$$

Interval: $x: [0, 4]$ and $y: [0, 5]$

$$m(0, 0) = 0 - 0 + 0 + 0 = 0$$

$$m(0, 5) = -0 - 25 + 0 + 30 = 5$$

$$m(4, 0) = -16 - 0 + 16 + 0 = 0$$

$$m(4, 5) = -16 - 25 + 16 + 30 = 5$$

∴ Max: 13 at $m(2, 3)$, Min: 0 at (0, 0) and (4, 0)

2. $g(x, y) = xy + x^2 - 4y + 6$ interval $x: [-2, 2]$ & $y: [1, 3]$

→ Here:

Gradient:

$$\frac{\partial g}{\partial x} = y + 2x, \quad \frac{\partial g}{\partial y} = x - 4$$

$$\nabla g = \begin{bmatrix} y + 2x \\ x - 4 \end{bmatrix}$$

let gradient = 0

$$y + 2x = 0 \Rightarrow y = -2x$$

$$x - 4 = 0 \Rightarrow x = 4$$

since 4 doesn't belong on interval.

Interval $x = (-2, 2)$ and $y = (1, 3)$

$$g(-2, 1) = (-2)(1) + 4 - 4 + 6 = -2 + 4 - 4 + 6 = 4$$

$$g(-2, 3) = -6 + 4 - 12 + 6 = -8$$

$$g(2, 1) = 2 + 4 - 4 + 6 = 8$$

$$g(2, 3) = 6 + 4 - 12 + 6 = 4$$

$\therefore$ Maximum: $g(-2, 3)$

$\therefore$ Maximum: $g(2, 1) = 8$

Minimum: $g(-2, 3) = -8$

3) $p(x, y) = x^3 - 3xy + 4$ Interval: $x [0, 2], y [0, 3]$

- Here,

Gradient:

$$\frac{\partial p}{\partial x} = 3x^2 - 3y, \quad \frac{\partial p}{\partial y} = -3x$$

$$\nabla p = \begin{bmatrix} 3x^2 - 3y \\ -3x \end{bmatrix}$$

Set gradient = 0

$$-3x = 0 \Rightarrow x = 0$$

$$3x^2 - 3y = 0 \Rightarrow y = x^2 = 0$$

$$p(0, 0) = 4$$

$$p(2, 0) = 8 - 0 + 4 = 12$$

$$p(2, 3) = 8 - 18 + 4 = -6$$

$\therefore$ Maximum: $p(2, 0) = 12$

Minimum: $p(2, 3) = -6$

4. $h(x, y) = 2x^2 + 3y^2 - 12x - 18y + 20$ interval $x \in [1, 5]$ $y \in [0, 4]$

→ Here;

Gradient:

$$\frac{\partial h}{\partial x} = 4x - 12, \quad \frac{\partial h}{\partial y} = 6y - 18$$

Set gradient = 0

$$4x - 12 = 0 \Rightarrow x = 3,$$

$$6y - 18 = 0 \Rightarrow y = 3$$

$$h(3, 3) = 18 + 27 - 36 - 54 + 20 = -25$$

$$h(1, 0) = 2 - 12 + 0 - 0 + 20 = 10$$

$$h(5, 0) = 50 - 60 + 0 - 0 + 20 = 10$$

$$h(1, 4) = 2 - 12 + 48 - 72 + 20 = -14$$

$$h(5, 4) = 50 - 60 + 48 - 72 + 20 = -14$$

∴ Max: 10 at (1, 0) and (5, 0) And Min: -25 at (3, 3)

5. $f(x, y) = x^2 + y^2 - 4x - 6y + 13$ interval $x \in [1, 3]$ $y \in [1, 4]$

→ Here;

Gradient:

$$\frac{\partial f}{\partial x} = 2x - 4, \quad \frac{\partial f}{\partial y} = 2y - 6$$

Set gradient = 0

$$2x - 4 = 0 \Rightarrow x = 2,$$

$$2y - 6 = 0 \Rightarrow y = 3$$

$$f(2, 3) = 4 + 9 - 8 - 18 + 13 = 0$$

Intervals:

$$f(1, 1) = 1 + 1 - 4 - 6 + 13 = 5$$

$$f(1, 4) = 1 + 16 - 4 - 24 + 13 = 2$$

$$f(3, 1) = 9 + 1 - 12 - 6 + 13 = 5$$

$$f(3, 4) = 9 + 16 - 12 - 24 + 13 = 2$$

∴ Max: 5 at (1, 1) and (3, 1) Min: 0 at (2, 3)

1. System with Two Variables:

$$c) \begin{cases} 3x + 2y = -2 \\ x + 4y = 6 \end{cases}$$

- Here;

Given;

$$3x + 2y = -2 \rightarrow \text{eq}^{\text{th}} \text{ci})$$

$$x + 4y = 6 \rightarrow \text{eq}^{\text{th}} \text{cii})$$

now,

Converting into Matrix form

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}, \quad x = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} -2 \\ 6 \end{bmatrix} \text{ which}$$

$$\text{is } AX = B$$

$$\text{Determinant of } A (|A|) = 3 \times 4 - 1 \times 2 = 10 \neq 0$$

we know,

$$A^{-1} = \frac{1}{|A|} \times \text{Adj of } A$$

$$= \frac{1}{10} \times \begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$$

$$\text{To find } X = A^{-1}B$$

$$= \frac{1}{10} \begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} -2 \\ 6 \end{bmatrix}$$

$$= \frac{1}{10} \begin{bmatrix} 4 \times (-2) + (-2) \times 6 \\ -1 \times (-2) + 3 \times 6 \end{bmatrix}$$

$$= \frac{1}{10} \begin{bmatrix} -8 - 12 \\ 2 + 18 \end{bmatrix}$$

$$= \frac{1}{10} \begin{bmatrix} -20 \\ 20 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{20}{10} \\ \frac{20}{10} \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

$$\therefore x = -2 \text{ and } y = 2 \text{ represented as } (x, y) = (-2, 2)$$

$$f) \begin{cases} 2x + 3y = 8 \\ 4x + 6y = 20 \end{cases}$$

→ Here;

$$2x + 3y = 8 \rightarrow \text{eq}^n \text{ci})$$

$$4x + 6y = 20 \rightarrow \text{eq}^n \text{cii})$$

now,

Converting into Matrix form.

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}, \quad x = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 8 \\ 20 \end{bmatrix} \text{ which}$$

$$\text{is } Ax = B$$

$$\text{Determinant of } |A| = 2 \times 6 - 4 \times 3 = 12 - 12 = 0$$

now the matrix does not have inverse

Therefore,

To check if the eqⁿ have solution:

Multiplying eqⁿ ci) with 2

$$(2x + 3y = 8) \times 2 \text{ we get,}$$

$$4x + 6y = 16$$

$$\text{But eq}^n \text{cii) is: } 4x + 6y = 20$$

Since $16 \neq 20$, the eqⁿ cannot both be true

Hence, no solutions.

2. System with Three Variables:

$$e) \begin{cases} 5x + 3y + 9z = -1 \\ -2x + 3y - 2z = -2 \\ -x - 4y + 5z = 1 \end{cases}$$

→ Here;

$$5x + 3y + 9z = -1 \rightarrow \text{eq}^n \text{ci})$$

$$-2x + 3y - 2z = -2 \rightarrow \text{eq}^n \text{cii})$$

$$-x - 4y + 5z = 1 \rightarrow \text{eq}^n \text{ciii})$$

Using Gauss Elimination method:

Converting into Matrix Form,

$$A = \begin{bmatrix} 5 & 3 & 9 \\ -2 & 3 & -1 \\ -1 & -4 & 5 \end{bmatrix}, \quad x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$$

now converting in augmented matrix,

$$\left[\begin{array}{ccc|c} 5 & 3 & 9 & -1 \\ -2 & 3 & -1 & -2 \\ -1 & -4 & 5 & 1 \end{array} \right] \text{ which is } [A|B]$$

then,

Pivot 1 (Dividing R_1 by 5)

$$R_1 \rightarrow R_1 / 5$$

$$\left[\begin{array}{ccc|c} 1 & 3/5 & 9/5 & -1/5 \\ -2 & 3 & -1 & -2 \\ -1 & -4 & 5 & 1 \end{array} \right]$$

now,

$$R_2 = R_2 + 2R_1, \quad R_3 = R_3 + R_1$$

$$\left[\begin{array}{ccc|c} 1 & 3/5 & 9/5 & -1/5 \\ 0 & 13/5 & 17/5 & -13/5 \\ 0 & -17/5 & 34/5 & 4/5 \end{array} \right]$$

again,

making pivot in 2nd row

$$R_2 = R_2 \times 5$$

$$\left[\begin{array}{ccc|c} 1 & 3/5 & 9/5 & -1/5 \\ 0 & 1 & 13/5 & -13/5 \\ 0 & -17/5 & 34/5 & 4/5 \end{array} \right]$$

Eliminating second column in R_1 and R_3

$$\left[\begin{array}{ccc|c} 1 & 3/5 & 9/5 & -1/5 \\ 0 & 1 & 13/5 & -13/5 \\ 0 & 0 & 95/5 & -40/5 \end{array} \right]$$

Then,

$$\frac{95}{21} z = -\frac{40}{35}$$

$$\text{or, } z = \frac{-8}{7} \times \frac{21}{95} = \frac{-168}{665}$$

Finding the value of y

$$y + \frac{13}{21} z = \frac{-4}{7}$$

$$\text{or, } y = \frac{-4}{7} + \frac{13 \times \frac{-168}{665}}{21}$$

$$\text{or, } y = \frac{-4}{7} + \frac{2184}{13965} = \frac{-21851}{13965} \times \frac{-4}{7} = \frac{-7980}{13965} = \frac{-1932}{4655}$$

again,

$$x + \frac{3}{5} y + \frac{9}{5} z = -\frac{1}{5}$$

$$\text{or, } x + \frac{3}{5} \times \left(\frac{-1036}{10995} \right) + \frac{9}{5} \times \left(\frac{-8}{35} \right) = -\frac{1}{5}$$

$$\text{or, } x = \frac{659}{665}$$

Again,

$$x + \frac{3}{5} y + \frac{9}{5} z = -\frac{1}{5}$$

$$\text{or, } x + \frac{3}{5} \left(\frac{-1932}{4655} \right) + \frac{9}{5} \times \left(\frac{-168}{665} \right) = -\frac{1}{5}$$

$$\text{or, } x + \left(\frac{-5796}{23275} \right) + \left(\frac{-1512}{3325} \right) = -\frac{1}{5}$$

$$\text{or, } x = -\frac{4655}{23275} + \frac{16380}{23275}$$

$$x = \frac{11725}{23275} \approx 0.504$$

$$\therefore x = \frac{11725}{23275} \approx 0.504, \quad y = \frac{-1932}{4655} \approx -0.415$$

$$\text{and } z = \frac{-168}{665} \approx -0.252$$

$$f) \begin{cases} 6x - 5y + 2z = 3 \\ 2x + y - 4z = 5 \\ 3x - 3y + 2z = -1 \end{cases}$$

- Here;

$$6x - 5y + 2z = 3 \rightarrow \text{eq}^n \text{ci})$$

$$2x + y - 4z = 5 \rightarrow \text{eq}^n \text{cii})$$

$$3x - 3y + 2z = -1 \rightarrow \text{eq}^n \text{ciii})$$

Converting into Matrix Form

$$A = \begin{bmatrix} 6 & -5 & 2 \\ 2 & 1 & -4 \\ 3 & -3 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, Y = \begin{bmatrix} 3 \\ 5 \\ -1 \end{bmatrix}$$

then,

Determinant of A (|A|) =

starting with augmented matrix

$$[A|I] = \left[\begin{array}{ccc|ccc} 6 & -5 & 2 & 1 & 0 & 0 \\ 2 & 1 & -4 & 0 & 1 & 0 \\ 3 & -3 & 2 & 0 & 0 & 1 \end{array} \right]$$

making pivot 1 in R_1

$$R_1 \rightarrow \frac{R_1}{6}$$

$$\left[\begin{array}{ccc|ccc} 1 & -5/6 & 1/3 & 1/6 & 0 & 0 \\ 2 & 1 & -4 & 0 & 1 & 0 \\ 3 & -3 & 2 & 0 & 0 & 1 \end{array} \right]$$

eliminating 1st column in R_2 and R_3

$$R_2 \rightarrow R_2 - 2R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & -5/6 & 1/3 & 1/6 & 0 & 0 \\ 0 & 16/6 & -14/3 & -1/3 & 1 & 0 \\ 3 & -3 & 2 & 0 & 0 & 1 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & -5/6 & 1/3 & 1/6 & 0 & 0 \\ 0 & 1/6 & -1/3 & -1/3 & 1 & 0 \\ 0 & -3/2 & 0 & -1/2 & 0 & 1 \end{array} \right]$$

making pivot in $R_2 = 1$

$$R_2 \rightarrow R_2 \times \frac{6}{1/6}$$

$$\left[\begin{array}{ccc|ccc} 1 & -5/6 & 1/3 & 1/6 & 0 & 0 \\ 0 & 1 & -2/3 & -1/3 & 6 & 0 \\ 0 & -3/2 & 0 & -1/2 & 0 & 1 \end{array} \right]$$

eliminating colm 2 in R_1 and R_3

$$R_1 \rightarrow R_1 + \frac{5}{6} R_2 \quad \text{and} \quad R_3 \rightarrow R_3 + \frac{3}{2} R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1/6 & 1/2 & 5/6 & 0 \\ 0 & 1 & -2/3 & -1/3 & 6 & 0 \\ 0 & 0 & -2/3 & -1/2 & 9 & 1 \end{array} \right]$$

making pivot in $R_3 = 1$

$$R_3 = R_3 \times \frac{-3}{-2/3} = [0, 0, 1 | \frac{16}{48}, \frac{-24}{21}, \frac{-16}{21}]$$

eliminating colm 3 in R_1 and R_2

$$R_1 = R_1 + \frac{1}{6} R_3$$

$$R_2 = R_2 + \frac{2}{3} R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5/48 & 23/112 & -1/21 \\ 0 & 1 & 0 & 1/6 & -5/8 & -2/3 \\ 0 & 0 & 1 & 1/3 & -3/7 & -16/21 \end{array} \right]$$

Thus

$$A^{-1} = \begin{bmatrix} \frac{5}{4} & \frac{23}{12} & -\frac{1}{2} \\ \frac{1}{6} & -\frac{5}{8} & -\frac{2}{3} \\ \frac{1}{3} & -\frac{8}{9} & -\frac{1}{6} \end{bmatrix}$$

Multiplying both sides of the matrix by A^{-1}

$$A^{-1}AX = A^{-1}B$$

$$\begin{bmatrix} \frac{5}{4} & \frac{23}{12} & -\frac{1}{2} \\ \frac{1}{6} & -\frac{5}{8} & -\frac{2}{3} \\ \frac{1}{3} & -\frac{8}{9} & -\frac{1}{6} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{5}{4} & \frac{23}{12} & -\frac{1}{2} \\ \frac{1}{6} & -\frac{5}{8} & -\frac{2}{3} \\ \frac{1}{3} & -\frac{8}{9} & -\frac{1}{6} \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ -1 \end{bmatrix}$$

→ (iii)

Solving by multiplying matrix in eq (iii)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{263}{168} \\ -2 \\ -4 \end{bmatrix}$$

$$\therefore x = -\frac{263}{168} \approx 1.566$$

$$y = -2$$

$$z = (-4) \times 5 = (-20)$$